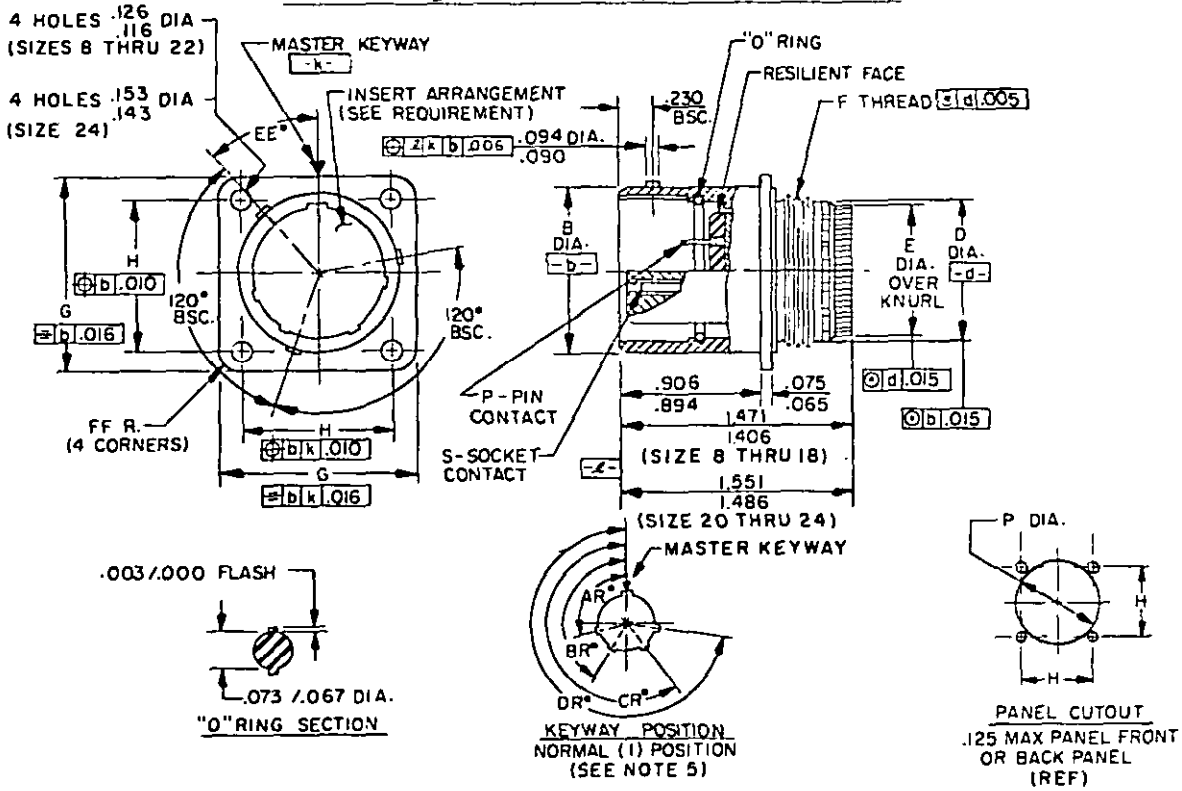
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISSCONNECT, ENVIRONMENT RESISTING; AND ACCESSORIES: RECEPTACLE, FLANGE MOUNT, CRIMP-TYPE CONTACTS, CLASS A, F AND E (SERIES 1)

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM	INCHES	MM
.003	.08	.116	2.95
.005	.13	.125	3.18
.006	.15	.126	3.20
.010	.25	.200	5.08
.015	.38	.230	5.84
.016	.41	.894	22.71
.065	1.65	.906	23.01
.067	1.70	1.406	35.71
.073	1.85	1.436	36.47
.075	1.91	1.471	37.36
.090	2.29	1.531	38.89
.094	2.39	.143	3.63
		.153	3.89

FSC 5935

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TABLE I.

Shell size	B dia	D dia	E dia	F thread	G	H	P dia	AR[1] BSC	BR[2] BSC	CR[2] BSC	DR[2] BSC	FF rad	EE° BSC Position 1	2	3	4	5	6
8	.553 (14.05)	.430 (10.92)	.422 (10.72)	½-28UNEF-2A	.820 (20.83)	.599 (15.21)	.640 (16.26)	105	140	215	265	.120 (3.05)	60	46	43	60	89	43
	.547 (13.89)	.424 (10.77)	.407 (10.34)		.804 (20.42)	.589 (14.96)	.630 (16.00)					.100 (2.54)						
10	.678 (17.22)	.555 (14.10)	.548 (13.92)	5/8-28UN-2A	.945 (24.00)	.724 (18.39)	.765 (19.43)	95	141	208	236	.120 (3.05)	53	87	51	77	87	87
	.672 (17.07)	.549 (13.94)	.533 (13.54)		.929 (23.60)	.714 (18.14)	.755 (19.18)					.100 (2.54)						
14	.928 (23.57)	.805 (20.45)	.798 (20.27)	7/8-28UN-2A	1.133 (28.78)	.911 (23.14)	1.015 (25.78)	95	141	208	236	.120 (3.05)	53	87	51	77	87	87
	.922 (23.42)	.799 (20.29)	.783 (19.89)		1.117 (28.37)	.901 (22.89)	1.005 (25.53)					.100 (2.54)						
16	1.054 (26.77)	.930 (23.62)	.924 (23.47)	1-28UN-2A	1.258 (31.95)	.974 (24.74)	1.140 (28.96)	80	142	196	293	.150 (3.81)	98	98	98	98	98	98
	1.048 (26.62)	.924 (23.47)	.909 (23.09)		1.242 (31.55)	.964 (24.49)	1.130 (28.70)					.130 (3.30)						
18	1.178 (29.92)	1.055 (26.80)	1.048 (26.62)	1-1/8-28UN-2A	1.351 (34.32)	1.067 (27.10)	1.265 (32.13)	80	142	196	293	.150 (3.81)	98	98	98	98	98	98
	1.172 (29.77)	1.049 (26.64)	1.033 (26.24)		1.335 (33.91)	1.057 (26.85)	1.255 (31.88)					.130 (3.30)						
20	1.303 (33.10)	1.180 (29.97)	1.173 (29.79)	1¼-28UN-2A	1.475 (37.47)	1.161 (29.49)	1.390 (35.31)	80	142	196	293	.150 (3.81)	98	98	98	98	98	98
	1.297 (32.94)	1.174 (29.82)	1.158 (29.41)		1.459 (37.06)	1.151 (29.24)	1.380 (35.05)					.130 (3.30)						
22	1.428 (36.27)	1.305 (33.15)	1.298 (32.97)	1-3/8-28UN-2A	1.570 (39.88)	1.255 (31.88)	1.515 (38.48)	80	142	196	293	.150 (3.81)	98	98	98	98	98	98
	1.422 (36.12)	1.299 (32.99)	1.283 (32.59)		1.554 (39.47)	1.245 (31.62)	1.505 (38.23)					.130 (3.30)						
24	1.553 (39.45)	1.430 (36.32)	1.423 (36.14)	1½-28UN-2A	1.711 (43.46)	1.380 (35.05)	1.640 (41.66)	80	142	196	293	.150 (3.81)	98	98	98	98	98	98
	1.547 (39.29)	1.424 (36.17)	1.408 (35.76)		1.695 (43.05)	1.370 (34.80)	1.630 (41.40)					.130 (3.30)						

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimensions are after plating.
5. Normal mating keyway position shown. For alternate keyway positions see MIL-C-81511, figure 8.

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Series: 1

Classes: A, F, and E.

Shell size: See table II.

Insert arrangement: See supplement for MS sheet corresponding to desired shell size.

Contacts: Style P or S, crimp type.

Shell polarization: 1 thru 6, key position.

Weight: See table II.

Marking: Part number on shell.

Mating counterpart: Plug connectors specified in MIL-C-81511/26 and /38.

Insertion and removal tool: As specified in MS3323, MS3342, and MS3344.

Part number: The part number shall consist of M81511/21 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions. Information concerning the formulation of the part number can be found in MIL-C-81511.

TABLE II.

Shell size	Weight (REF) 1/ (pounds)
24 (code J)	.122
22 (code H)	.109
20 (code G)	.095
18 (code F)	.080
16 (code E)	.067
14 (code D)	.056
10 (code B)	.035
8 (code A)	.027

1/ Removable crimp contacts not included in weight.
(See contact drawings for weight of pin and socket contact.)

Custodians:
Army - EL
Navy - AS
Air Force - 17

Review activities:
Army - MI, MU, AV, WC
Navy - SH, EC, OS
Air Force - 11, 80
DSA - ES

User activities:
Army - ME, AT
Navy - MC
Air Force -

Preparing activity:
Navy - AS

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